

Chemistry content

Content overview

Students are expected to demonstrate and apply the knowledge, understanding and skills described in the content. They are also expected to analyse, interpret and evaluate a range of scientific information, ideas and evidence using their knowledge, understanding and skills.

To demonstrate their knowledge, students should be able to undertake a range of activities, including the ability to recall, describe and define, as appropriate.

To demonstrate their understanding, students should be able to explain ideas and use their knowledge to apply, analyse, interpret and evaluate, as appropriate.

Students should develop their ability to apply mathematical skills to chemistry throughout the course. These skills include the ability to change the subject of an equation, substitute numerical values and solve algebraic equations using decimal and standard form, ratios, fractions and percentages. Further details of the skills that should be developed are given in *Appendix 6: Mathematical skills and exemplifications*. Students should also be familiar with *Système Internationale d'Unités* (SI) units and their prefixes, be able to estimate physical quantities and know the limits of physical measurements.

Practical work is central to any study of chemistry. For this reason, the specification includes 16 core practical activities and further suggested practicals, which form a thread linking theoretical knowledge and understanding to practical scenarios. In following this thread, students will build on practical skills learned at GCSE (or equivalent), becoming confident practical chemists, handling apparatus competently and safely. Using a variety of apparatus and techniques, they should be able to design and carry out both the core practical activities and their own investigations, collecting data which can be analysed and used to draw valid conclusions.

Questions in examination papers will aim to assess the knowledge and understanding that students gain while carrying out practical activities, within the context of the 16 core practical activities, as well as in novel practical scenarios. Success in questions that indirectly assess practical skills will come more naturally to those candidates who have a solid foundation of laboratory practice and who, having carried them out, have a thorough understanding of practical techniques. Therefore, where possible, teachers should consider adding additional experiments to the core practical activities.

Students will be assessed on their practical skills in papers 3 and 6. These papers will include testing the skills of students in planning practical work – both in familiar and unfamiliar applications – including risk management and the selection of apparatus, with reasons.

When data handling, students will be expected to use significant figures appropriately, to process data and to plot graphs.

Students should be encouraged to use information technology throughout the course.

Units

Unit 1: Structure, Bonding and Introduction to Organic Chemistry	16
Unit 2: Energetics, Group Chemistry, Halogenoalkanes and Alcohols	27
Unit 3: Practical Skills in Chemistry I	41
Unit 4: Rates, Equilibria and Further Organic Chemistry	44
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Unit 1: Structure, Bonding and Introduction to Organic Chemistry

IAS compulsory unit

Externally assessed

Unit description

Introduction

This unit gives students opportunities to develop the basic chemical skills of writing formulae and equations, and calculating chemical quantities.

The study of atomic structure includes a description of *s*, *p*, and *d* orbitals and shows how electronic configurations can account for the arrangement of elements in the Periodic Table. This leads to an appreciation of one of the central features of chemistry: the explanation of the properties of elements and the patterns in the Periodic Table in terms of atomic structure.

An understanding of the electronic structure of atoms leads to an appreciation of the three types of strong chemical bonding: ionic, covalent and metallic. Following from this, shapes of molecules can then be considered.

The basic principles of organic chemistry are covered and students study alkanes and alkenes, and will begin to develop a mechanistic approach to organic chemistry.

Chemistry in action

The study of atomic structure gives some insight into the methods that scientists use to study the structure of atoms. This leads to the introduction of the mass spectrometer and its importance in sensitive methods of analysis in areas such as space research, medical research and diagnosis, in detecting drugs in sport and in environmental monitoring.

Chemists set up theoretical models and gain insight by comparing real and theoretical properties of chemicals. This is illustrated in the unit by considering the evidence for the different kinds of chemical bonding.

Electron-pair repulsion theory is also used to show how chemists can develop theories and use them to make predictions.

Students start to use the conventions for mechanisms in organic chemistry as a way to represent the movement of electrons in reactions.

Practical skills

Students can start with simple test-tube reactions to illustrate a range of chemical equations. They can then build up to carrying out practical work that can be used to find reaction quantities, covered in the first core practical on molar volume.

Simple practical work can be used to investigate the properties of substances with different types of bonding.

The introduction to organic chemistry shows how chemists work safely with hazardous chemicals by managing risks. A number of possible practicals can be used to explore the chemistry of alkenes.

Mathematical skills

There are opportunities for the development of mathematical skills in this unit. This includes converting between units such as cm^3 and dm^3 , using standard form with the Avogadro constant, rearranging formulae for calculating moles in solids and in solutions and the ideal gas equation, calculating atom economy, dealing with percentage errors, calculating a relative atomic mass from isotopic composition data, using simple probability to calculate the peak heights for the mass spectrum of molecules such as chlorine, using logarithms to compare successive ionisation energies for an element, representing shapes of molecules with suitable sketches, plotting data to investigate trends in boiling temperatures of alkanes and calculating the yield of a reaction. (Please see *Appendix 6: Mathematical skills and exemplifications* for more detail.)

Assessment information

- First assessment: January 2019.
 - The assessment is 1 hour and 30 minutes.
 - The assessment is out of 80 marks.
 - Students must answer all questions.
 - This paper has two sections:
 - Section A: multiple choice questions
 - Section B: mixture of short-open, open-response and calculation questions.
 - This paper will include a minimum of 18 marks that target mathematics at Level 2 or above.
 - Students will be expected to apply their knowledge and understanding of experimental methods in familiar and unfamiliar contexts.
 - Calculators may be used in the examination (see *Appendix 8: Use of calculators*).
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